

Tutorial 0

Power & Energy Calculations

$$\rightarrow x(t) = A \sin(2\pi ft) = v(t)$$

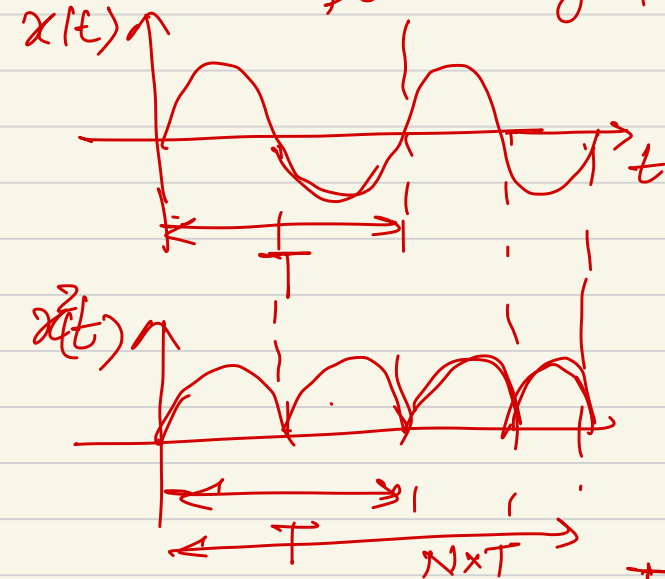
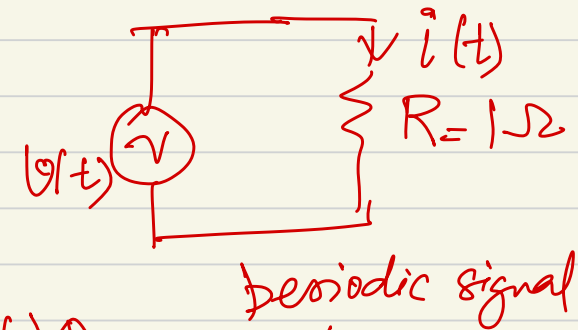
$$i(t) = \frac{v(t)}{R} = \frac{v(t)}{1\Omega}$$

$$p(t) = v(t) i(t) = x^2(t)$$

$$\text{Av. Power} = \frac{1}{T} \int_0^T x^2(t) dt$$

$$= \frac{1}{T} \int_0^T A^2 \sin^2(2\pi ft) dt$$

$$= \frac{A^2}{T} \int_0^T \underbrace{\frac{1 - \cos(2 \times 2\pi ft)}{2}}_{\text{trig identity}} dt = \frac{A^2}{2T} \left[t - \frac{\sin(4\pi ft)}{4\pi f} \right]_0^T$$
$$= \frac{A^2}{2T} \left[T - \frac{1}{4\pi f} \left[\sin\left(\frac{4\pi \times T}{T}\right) - \sin(0) \right] \right] = \frac{A^2}{2}$$



$$\text{average power} = \frac{A^2}{2}$$

Q1. $x_1(t) = \sin(2\pi 5t)$, $x_2(t) = 2\sin(2\pi 5t)$

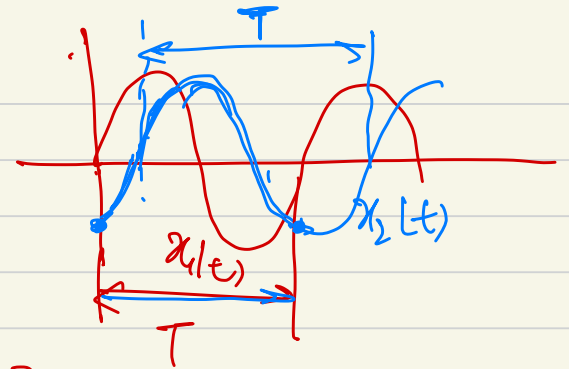
$$\begin{array}{ccc} A_1 = 1 & & A_2 = 2 \\ P_1 = \frac{1}{2} & < & P_2 = \frac{4}{2} = 2 \end{array}$$

$$E = P \times T_{\text{tot}} = P \times N \times T =$$

Q2. $x_1(t) = A \sin(\omega t)$
 $x_2(t) = A \sin(\omega t + \phi)$

$$P_1 = \frac{1}{T} \int_0^T x_1^2(t) dt, \quad P_2 = \frac{1}{T} \int_0^T x_2^2(t) dt$$

$$P = \frac{1}{T} \int_0^T (x_1(t) + x_2(t))^2 dt$$



$$= \frac{1}{T} \int_0^T [x_1^2(t) + x_2^2(t) + 2x_1(t)x_2(t)] dt$$

$$= \frac{1}{T} \left[\int_0^T x_1^2(t) dt + \int_0^T x_2^2(t) dt + 2 \int_0^T x_1(t)x_2(t) dt \right]$$

$$= \frac{1}{T} \left[\frac{A^2}{2} T + \frac{A^2}{2} T + 2A^2 \int_0^T \sin(\omega t) \sin(\omega t + \phi) dt \right]$$

$$= A^2 - \frac{2A^2}{T} \int_0^T \left[\frac{\cos(2\omega t + \phi) - \cos(\phi)}{2} \right] dt$$

$$= A^2 - \frac{A^2}{T} \left[\int_0^T \cos(2\omega t + \phi) dt - T \cos \phi \right]$$

$$= A^2 + A^2 \cos \phi - \frac{2A^2}{T} \left[\frac{\sin(2\omega t + \phi)}{2\omega} \right]_0^T$$

$$f = \frac{1}{T}$$

$$\omega = \frac{2\pi}{T}$$

$$= \quad // \quad - \frac{2A^2}{T} \cdot \frac{1}{2\omega} \left[\sin\left(2T \times \frac{2\pi}{T} + \phi\right) - \sin(\phi) \right]$$

$$= \quad // \quad - \frac{2A^2}{T} \cdot \frac{1}{2\omega} \left[\sin(4\pi + \phi) - \cancel{\sin(\phi)} \right]$$

$\Downarrow$
 ~~$\sin \phi$~~

$$P(\phi) = A^2 (1 + \cos \phi)$$

$$\phi = 0, \quad P_{\max} = 2A^2$$

$$\phi = \pi, \quad P_{\min} = 0$$

